

Regressão linear simples e correlação linear simples (Pearson)

$$SPD_{xy} = \sum_{i=1}^n (X_i \cdot Y_i) - \frac{\left(\sum_{i=1}^n X_i\right) \cdot \left(\sum_{i=1}^n Y_i\right)}{n}$$

$$SQD_y = \sum_{i=1}^n Y_i^2 - \frac{\left(\sum_{i=1}^n Y_i\right)^2}{n}$$

$$SQD_x = \sum_{i=1}^n X_i^2 - \frac{\left(\sum_{i=1}^n X_i\right)^2}{n}$$

$$\hat{\beta}_1 = \frac{SPD_{xy}}{SQD_x}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \cdot \bar{X}$$

$$e_{ij} = Y_{ij} - \hat{Y}_{ij}$$

$$R^2 = \frac{SQRegressão}{SQTotal}$$

$$R^2 = \frac{\hat{\beta}_1 \cdot SPD_{xy}}{SQD_y}$$

$$R^2 = \frac{SPD_{xy}^2}{SQD_y \cdot SQD_x}$$

$$R^2 = (\hat{\beta}_1)^2 \cdot \frac{SQD_x}{SQD_y}$$

$$SQRegressão = \hat{\beta}_1 \cdot SPD_{xy} \quad SQRegressão = \frac{(SPD_{xy})^2}{SQD_x} \quad \rho_{xy} = \frac{SPD_{xy}}{\sqrt{SQD_x \cdot SQD_y}} \quad \rho_{xy}^2 = R^2$$

$$SQRegressão = \hat{\beta}_0 \cdot \sum_{i=1}^n Y_i + \hat{\beta}_1 \cdot \sum_{i=1}^n (Y_i \cdot X_i) - \frac{\left(\sum_{i=1}^n Y_i\right)^2}{n} \quad SQTotal = \sum_{i=1}^n Y_i^2 - \frac{\left(\sum_{i=1}^n Y_i\right)^2}{n}$$